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HEALTHCARE CHATBOT AND VIRTUAL FIRST AID GUIDE

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ABSTRACT

A good and healthy lifestyle depends heavily on healthcare. AI is now being utilized more and more in medical treatment. In order to offer automated online advice and healthcare support for patients, the concept of the healthcare chatbot is employed to develop and simulate human-machine dialogue. This chatbot uses AI's Machine Learning and NLP (Natural Language Processing) disciplines to identify the input words and phrases and provide question judgments for patient answers. One of the emerging technologies that will continue to hit the market as demand for machine learning and artificial intelligence (AI) rises is virtual assistant bots, commonly referred to as chatbots. Chatbots have now fully evolved from menu/button-based to keyword-based to contextual-based. Contextual-based chatbots are the most cutting-edge of all of the aforementioned since they employ machine learning and artificial intelligence techniques to store and analyze the training models that enable the chatbot to provide better and more suitable answers when users ask it questions about certain domains. Before contacting a doctor, the goal is to develop a medical chatbot that can identify the illness and offer basic information about it. The use of medical chatbots will improve access to medical knowledge while decreasing healthcare costs. Computer programs known as chatbots use natural language to converse with users. Our project's main goal is to give consumers quick, precise illness predictions based on symptoms. We have utilized the Decision tree algorithm to forecast illnesses. By offering a predicted diagnosis, chatbots have the potential to significantly alter the healthcare sector.

I. INTRODUCTION

An intelligent conversational software program called a chatbot simulates a text-based automated conversational interaction between humans and computers. The chatbot analyzes and evaluates patient input to provide an immediate, accurate response. The relational database management system (RDBMS) stores the conversation pattern's input phrase. When patients cannot immediately contact physicians and doctors, this technique is designed to assist them save money and time while in the hospital. It offers patients individualized diagnoses. Python is the programming language that was used to create this chatbot. Humans have been working nonstop for the past few decades, yet they frequently neglect to put their health first. Long-term, this issue puts people's quality of life in danger. However, thanks to artificial intelligence, we can now provide people with healthcare services whenever it's convenient for them and at a fair price. A healthy body is one of the greatest benefits we can have. Each of us aspires to a healthy physique and an improved quality of life. This paper's main objective is to offer these services in order to achieve the stated goal. High-tech devices have mostly taken over our life and it is impossible to envision a time without them. Artificial intelligence is a developing field as a result of its many research-related applications. Forecasting illnesses improves the accuracy of risk classification using vast amounts of data, which is one of the main goals of academics utilizing big data analytics. In general, e-healthcare facilities are an essential resource for developing countries, but setting them up might be difficult owing to inadequate infrastructure and understanding. Many internet users rely on it to discover solutions to their healthcare-related issues. We have developed a platform that allows users to access online medical services with the goal of supporting medical professionals.

II. LITERATURE REVIEW

1.1. “Multilingual Healthcare Chatbot Using Machine Learning” published by “Sagar Badlani, Tanvi Aditya” in May 2021.

Nowadays, the healthcare industry is one of the main focal areas in the globe. Health issues are surfacing more frequently. India has a significant management difficulty when it comes to rural healthcare. Early illness

detection and treatment can be extremely helpful. Physical consultation is expensive and time-consuming, especially in remote locations. Using healthcare chatbots is the answer. The suggested solution is a multilingual healthcare chatbot program that can diagnose diseases based on user symptoms. Additionally, it answers to user inquiries by comparing sentences using the TF-IDF and Cosine Similarity algorithms, then selecting the best answer from its knowledge store. The chatbot system's multilingual capabilities make it ideally suited for usage in rural India. English, Hindi, and Gujarati are the three languages that the chatbot program presently supports. The chatbot program uses natural language processing principles to have conversations with the user and also supports speech-to-text and text-to-speech conversion, allowing voice communication. For the purpose of predicting diseases, five alternative machine-learning algorithms have been examined. The Random Forest Classifier has the highest accuracy (98.43%) and offers the best results. As a result, it serves as the system's primary classifier.

1.2. "Health Chatbot Using Natural Language Processing for Disease Prediction and Treatment" published by "Philip Indra, Richard Aldy, Reinhart Perbowo Pujo Leksono" in October 2021.

A system that can answer common queries from those who are unaware of the goods or services offered by a corporation is necessary. The name of this system is Frequently Asked Questions. The information is not delivered by this system in an effective or efficient manner. They suggested a chatbot in this research that would use natural language processing to deliver certain health-related information. They create a chatbot system with NLP that can comprehend user inquiries and provide responses. In order to retrieve the responses from the document with the highest similarity, the cosine similarity is utilized to compare the query terms (the user's inquiries) and the documents. Based on our research, their medical chatbot accurately identified the user's disease with a success rate of about 87 percent.

Rationale

Hospitals are now able to give online healthcare help twenty-four hours a day because to this healthcare chatbot technology, which offers both detailed and general responses. Additionally, it aids in the creation of leads and automatically sends the data about those leads to sales. It aids in patient data tracking and offers clinicians insightful information. It offers a user-friendly design that prevents customers from becoming overwhelmed by information and guarantees they receive responses swiftly.

Feasibility Study

Technical Feasibility:

Technical viability considers the accessibility of the hardware and software needed for system creation, the maturity and compatibility of the technology suggested for usage, and the accessibility of the technical people needed to create the system. After the investigation, we came to the conclusion that we should keep using the tools and development environment that we had already selected. This was crucial in our situation since we were working on two distinct departmental phases that will eventually need to be linked to create a larger system.

Operational Feasibility:

The main focus of operational feasibility is potential operational issues. There are two facets to this matter:

- What is the likelihood that the developed solution won't be implemented or won't work?
- How favorable are the management and end users to the proposed solution?

Although there is a remote chance that management will reject the solution, there is a high likelihood that end users won't be eager to utilize it because of a lack of knowledge, insight, etc.

Economic Feasibility:

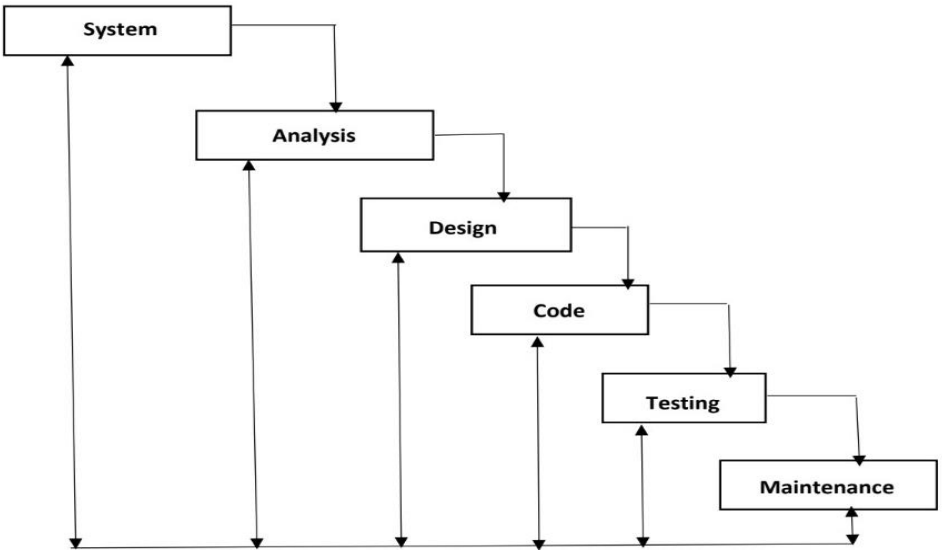
It serves as a gauge for the project's cost-effectiveness. Simply determining whether or not the potential benefits of fixing the problems are worthwhile is what economic feasibility is all about. It is difficult to estimate the cost at the feasibility study level since the needs of the members and potential solutions have not yet been determined. However, once the precise needs and solutions have been determined, the analyst does a "cost-benefit analysis" in which they compare all potential solutions' costs and advantages.

III. METHODOLOGY

This method uses the "Classic Life Cycle" approach because it is straightforward and works well for small-scale projects. System development life cycle (SDLC) is another name for the "Classic Life Cycle". The development of

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an information system via a number of distinguishable phases is how it is described. The SDLC is the collective name for these stages. Following is the breakdown of its stages that we employed in our project:



IV. SYSTEM DESIGN

System Architecture

The idea was The User Interface Component, User Message Analysis Component, Dialog Management Component, Backend Database, and Response Generation Component make up the chatbot system architecture. The user's spelling errors are corrected by a spell checker. The chatbot's NLU is recognized and translated using a machine translation.

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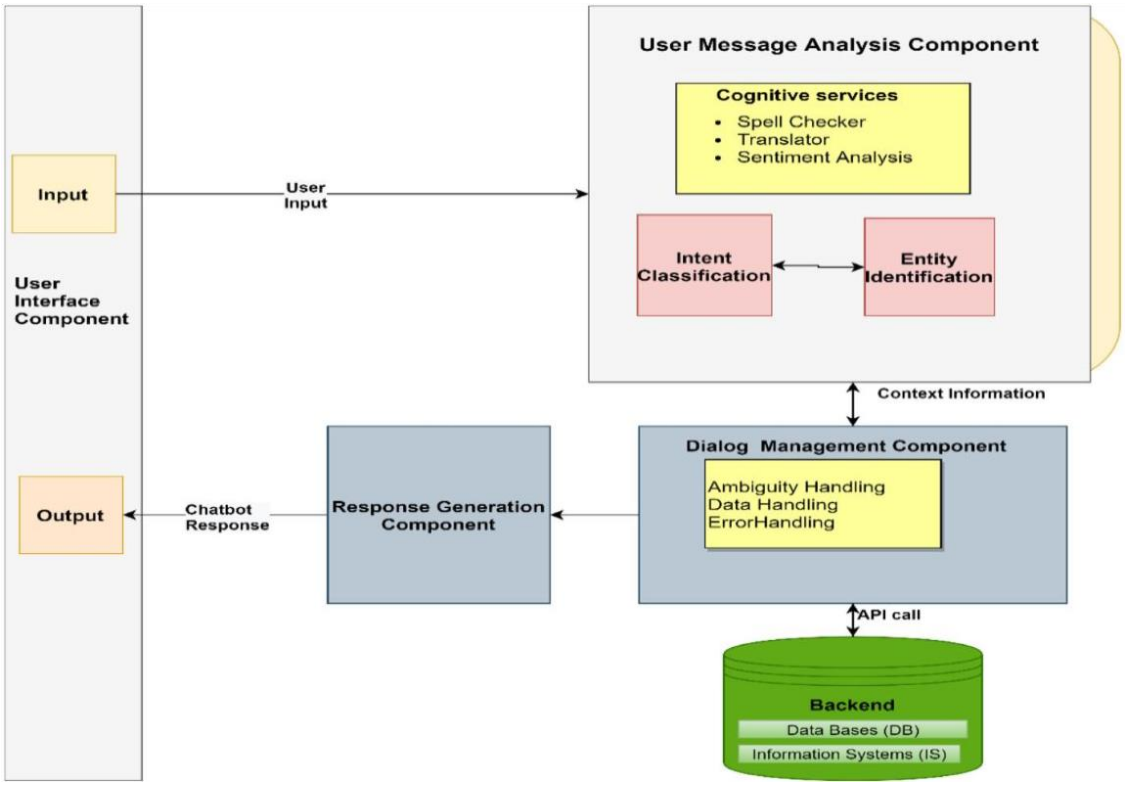


Figure1: System Architecture

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Modules

Admin module:

The administrator of this online application can edit and update the dataset that is used to forecast the results by logging in to the administrative page.

User module:

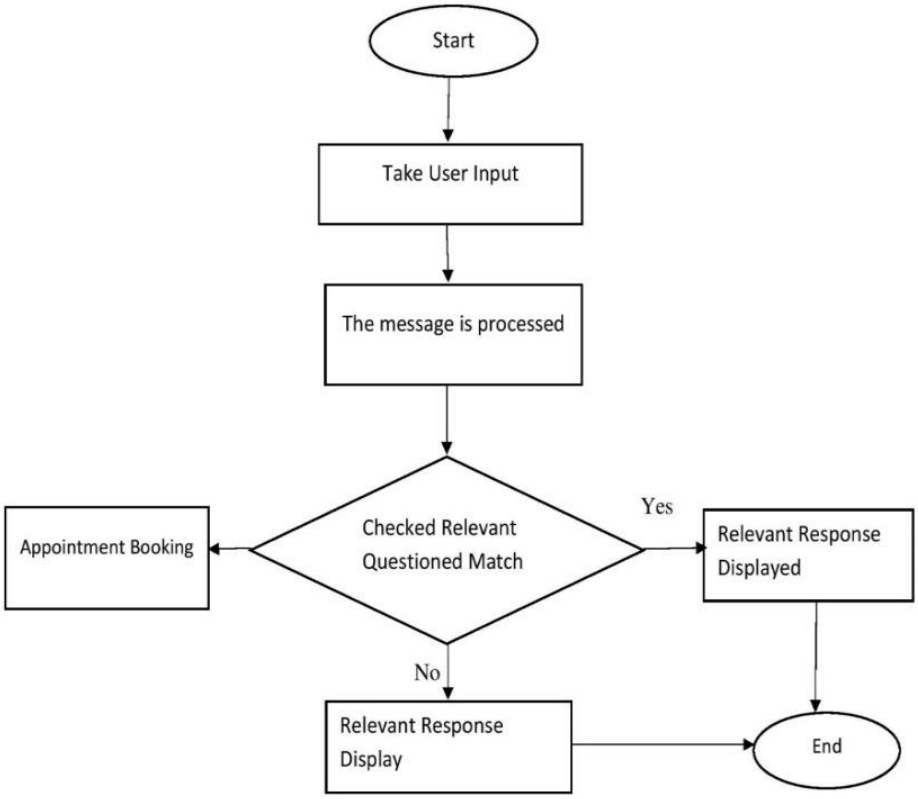
The user enters his or her basic information here to access the chatbot page. The user enters the details of their doubt, disease, or ailment, and the chatbot responds.

Doctor module:

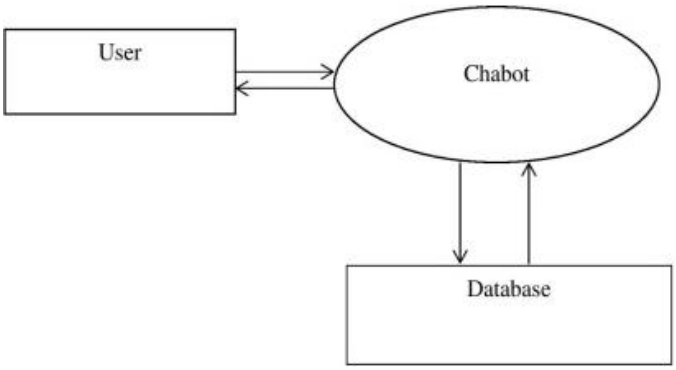
We offer the doctor login for doctors in this module, as well as information on whether a clinic or hospital is open or closed. View the associated scheduled appointment as well.

Data Flow Diagram

DFD Level 0

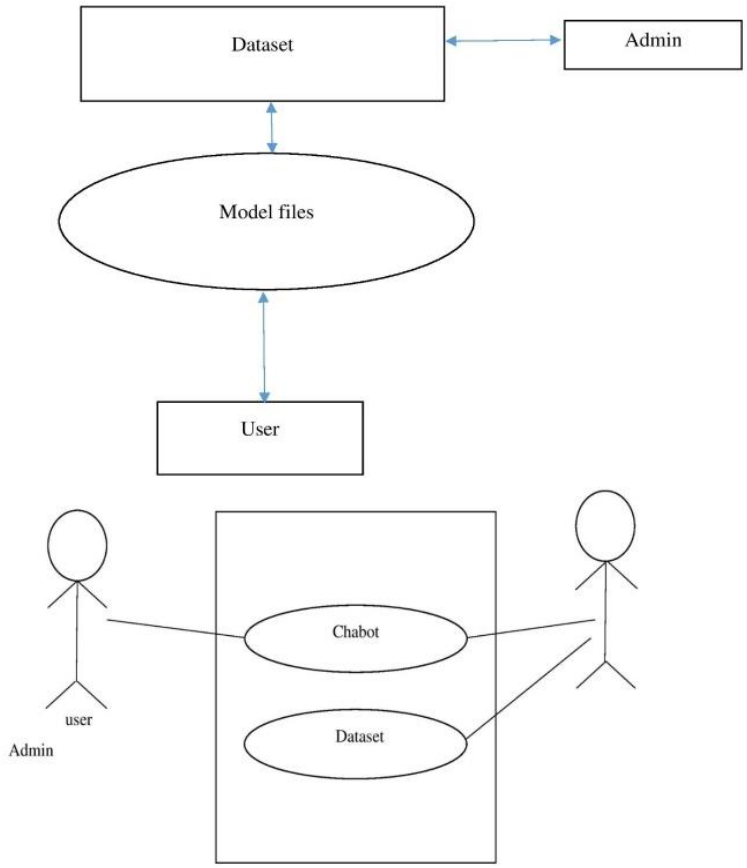


DFD Level 1



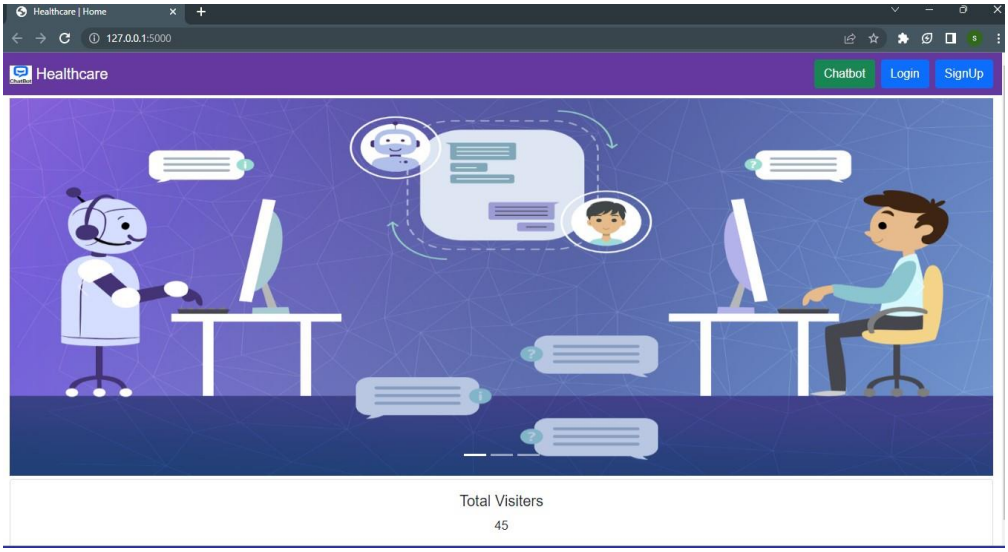
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DFD Level 2



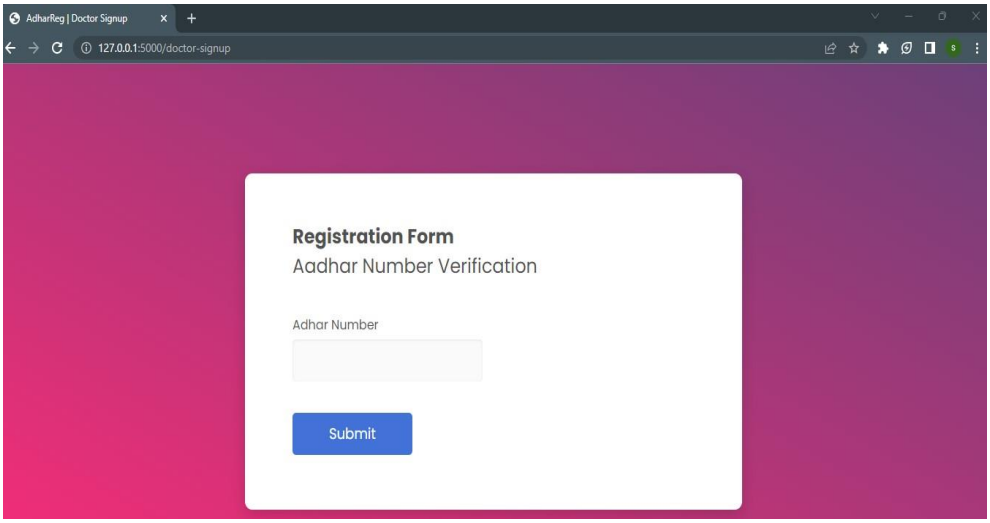
V. RESULTS

Home Page

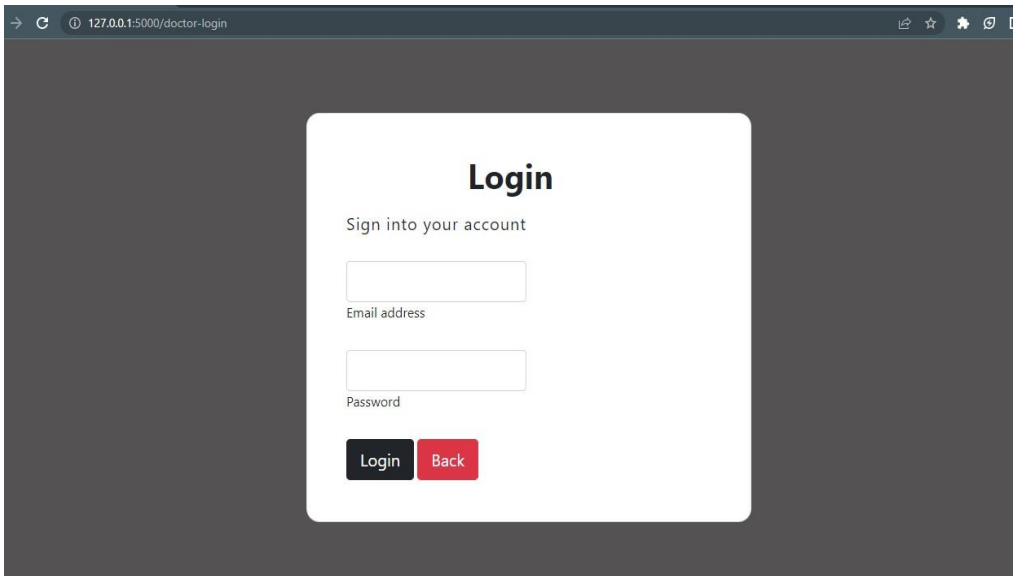


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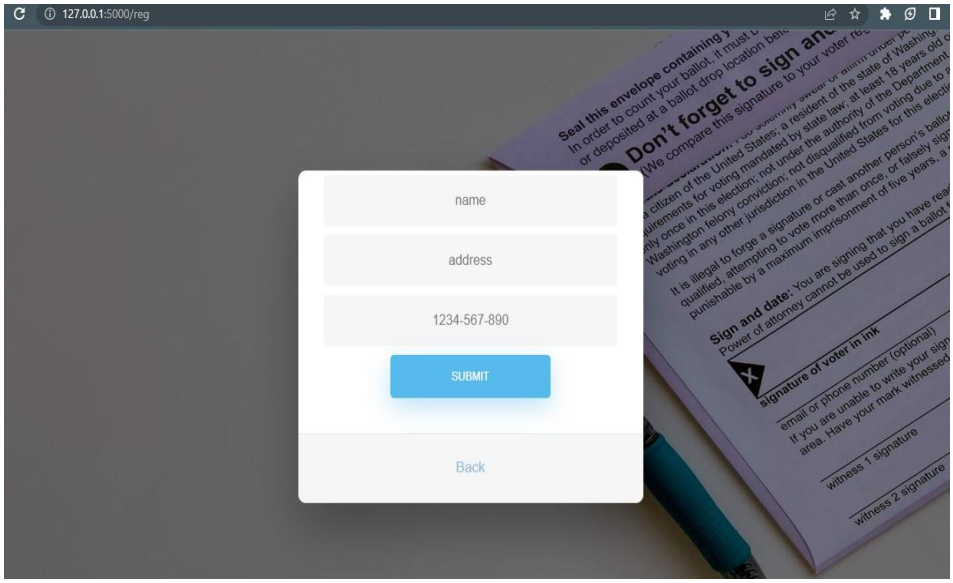
Registration Page



Login Page

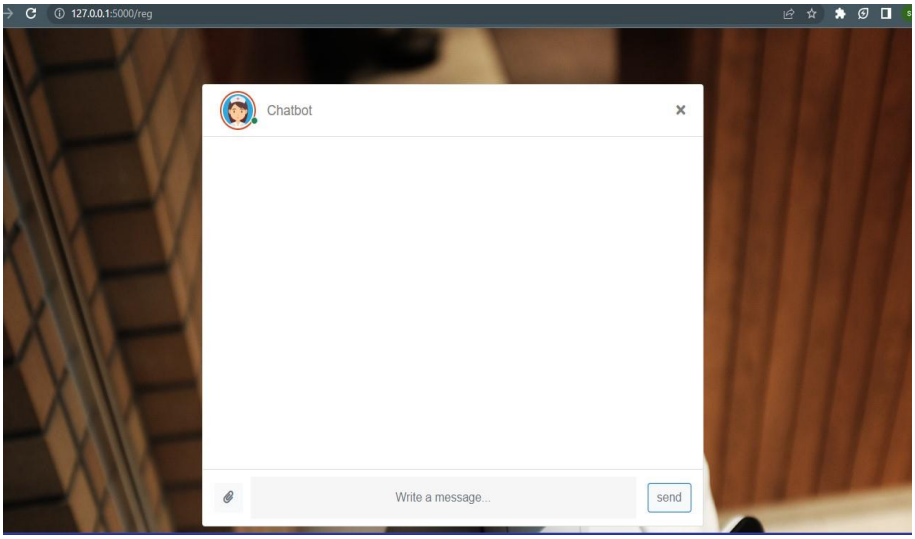


User Login

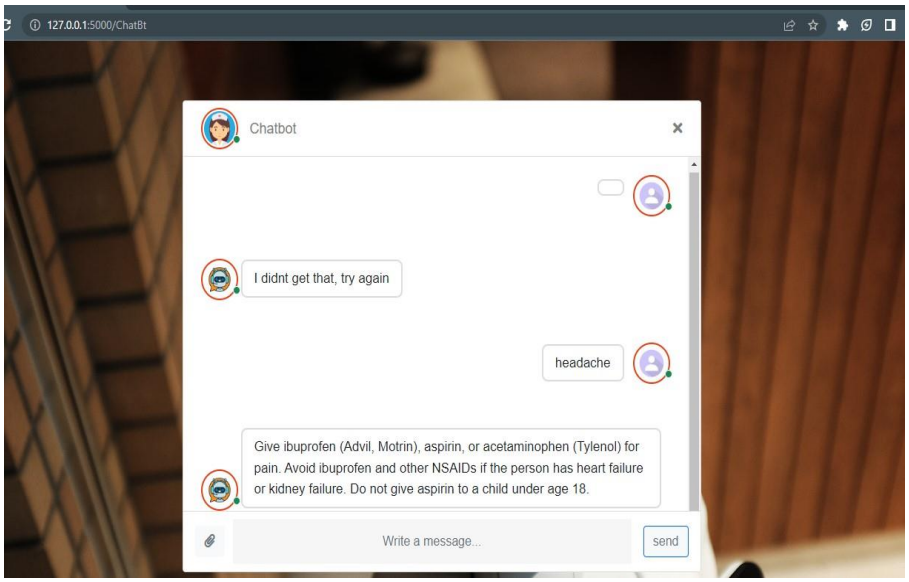


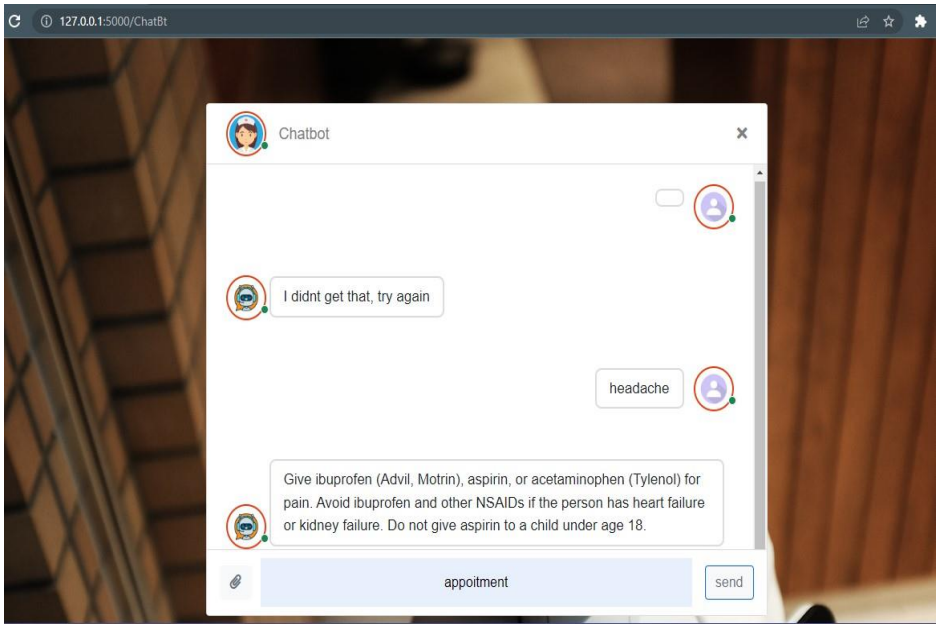
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Chatbot UI



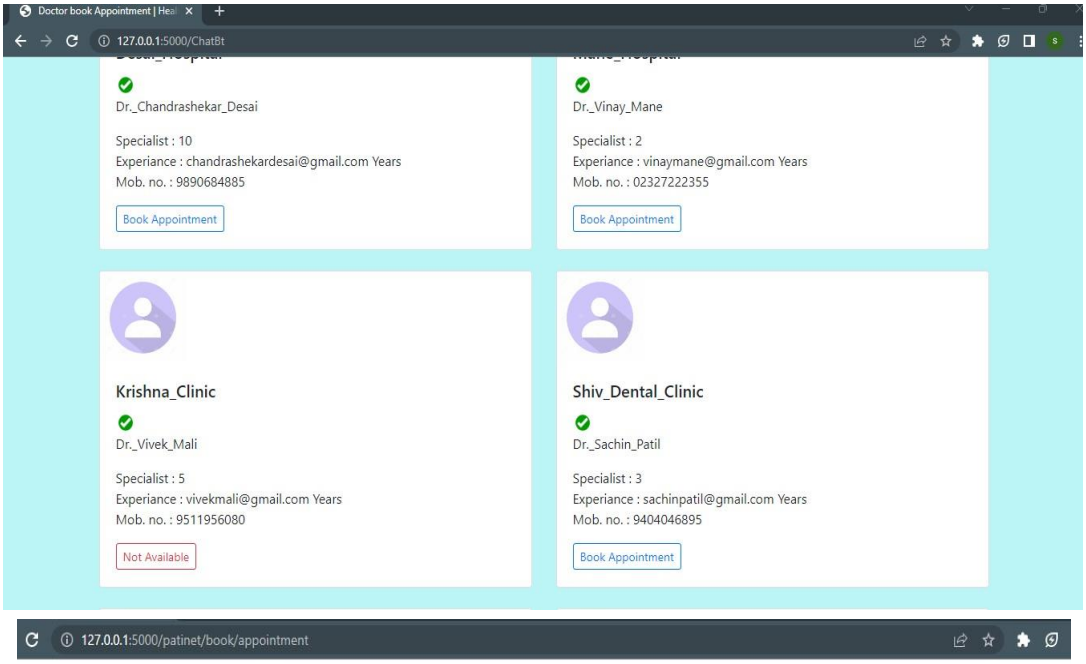
Chatbot Response

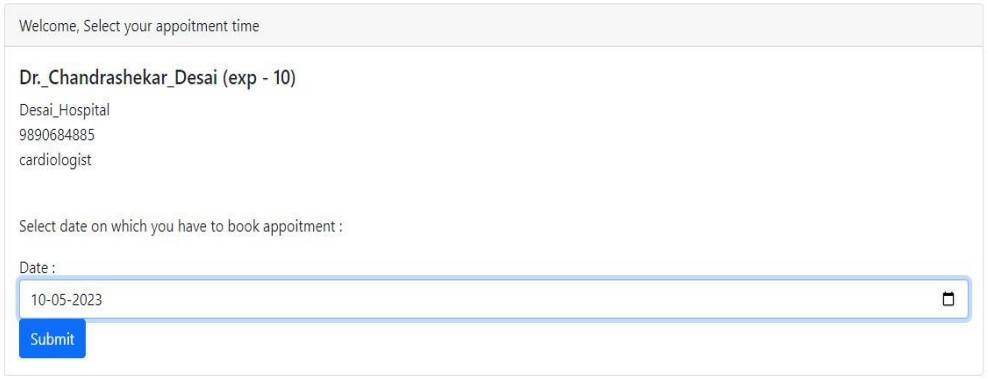




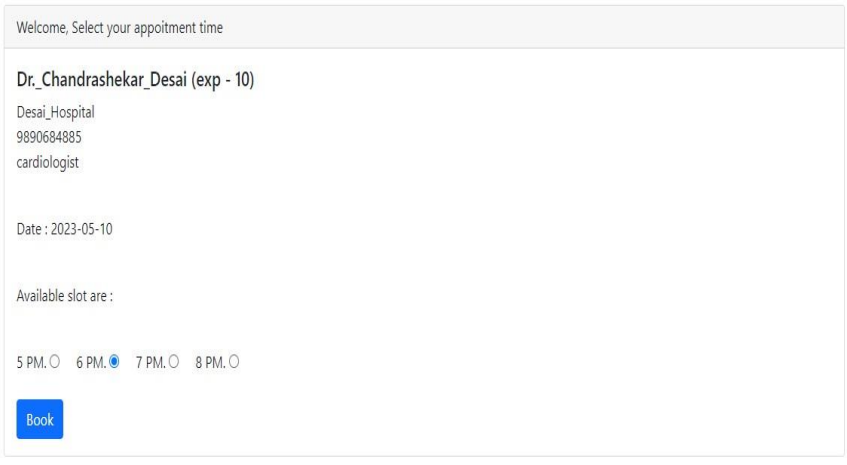
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Appointment Page



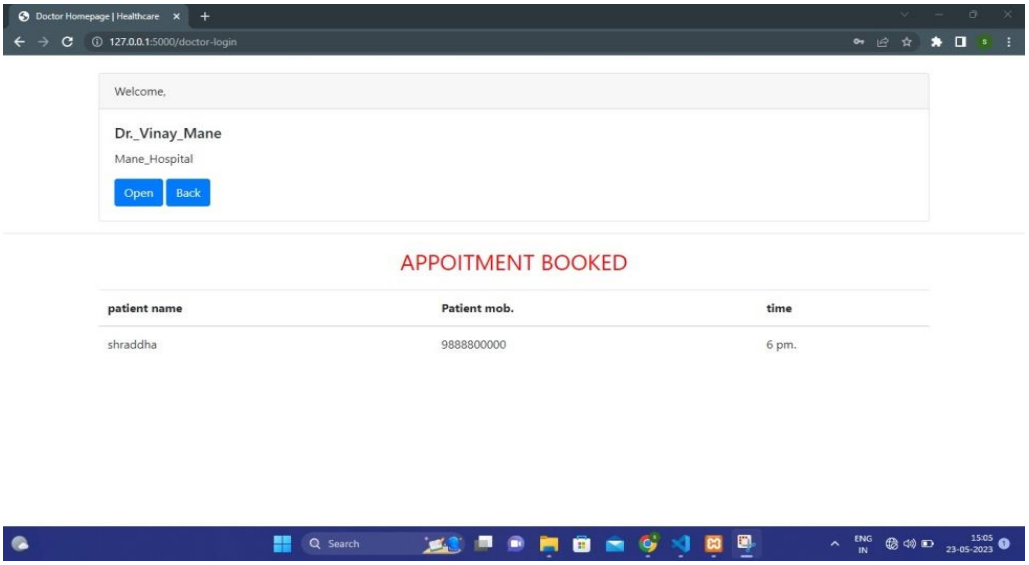




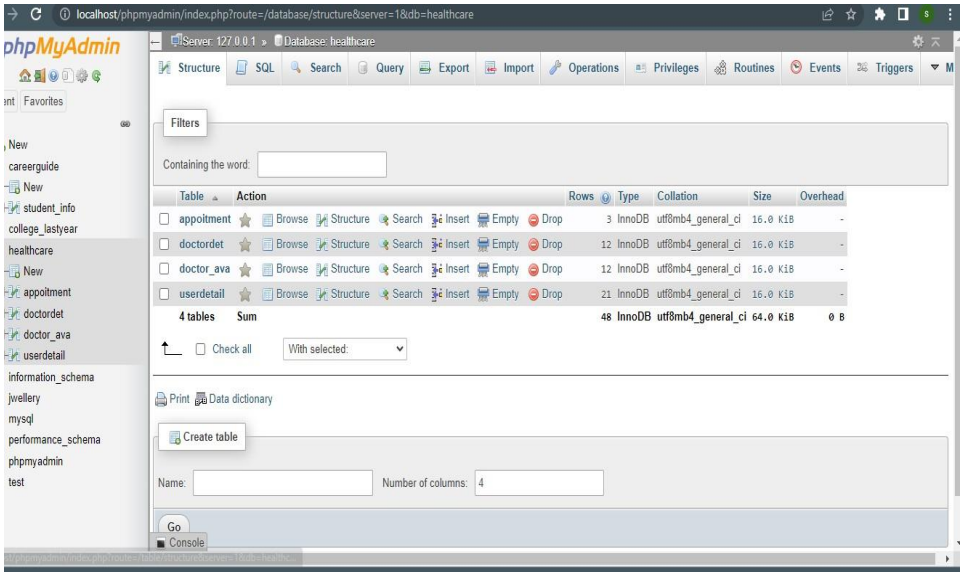


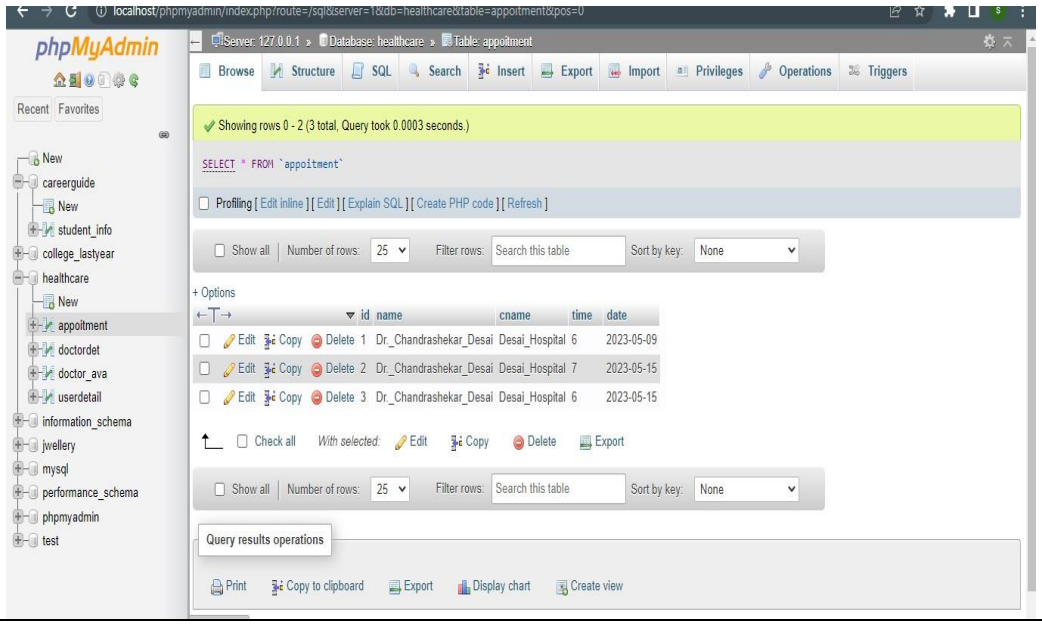
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Appointment Booked Page

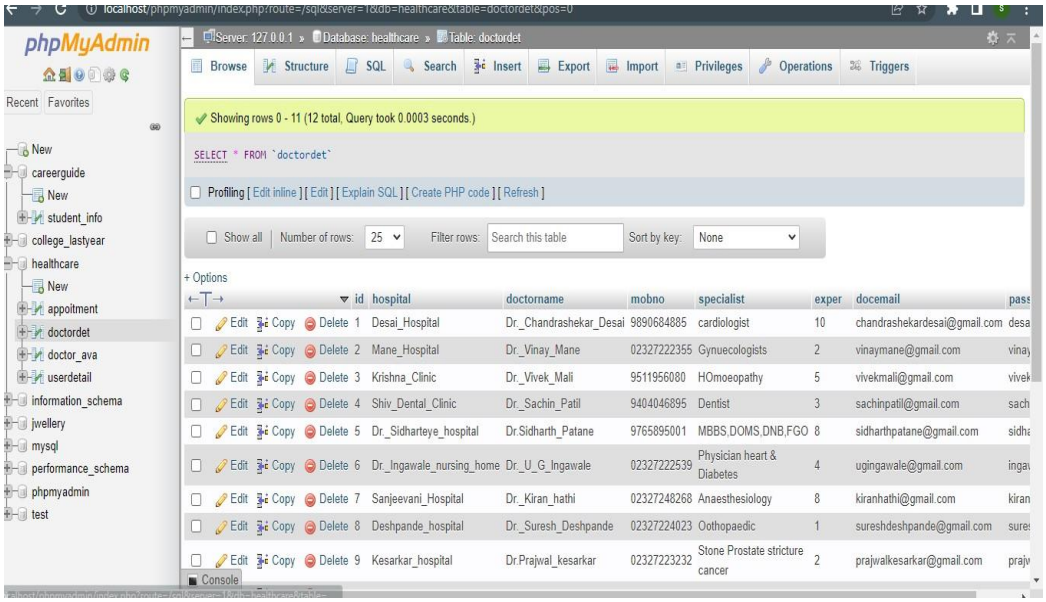


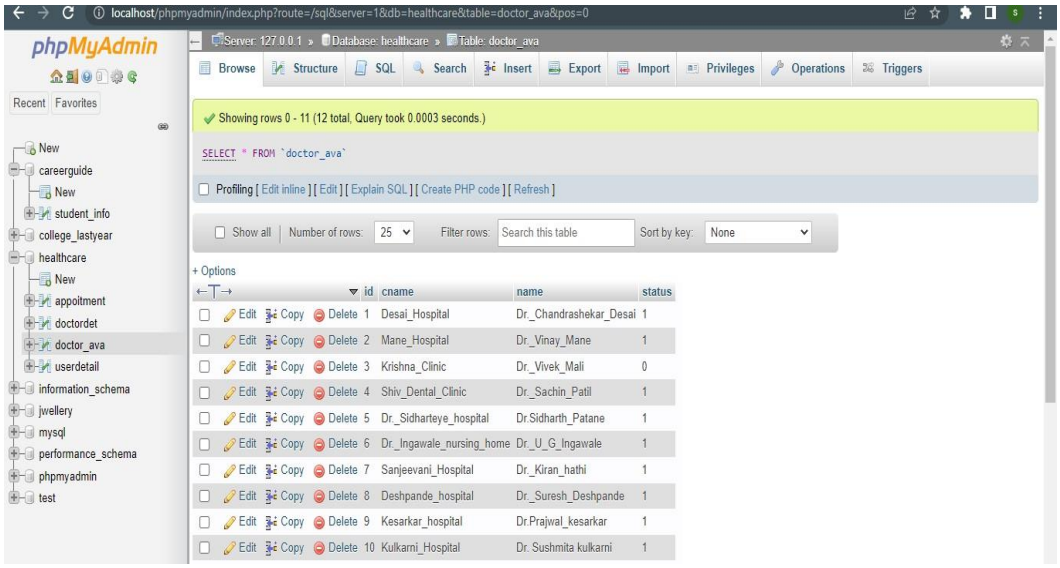
Database Pages

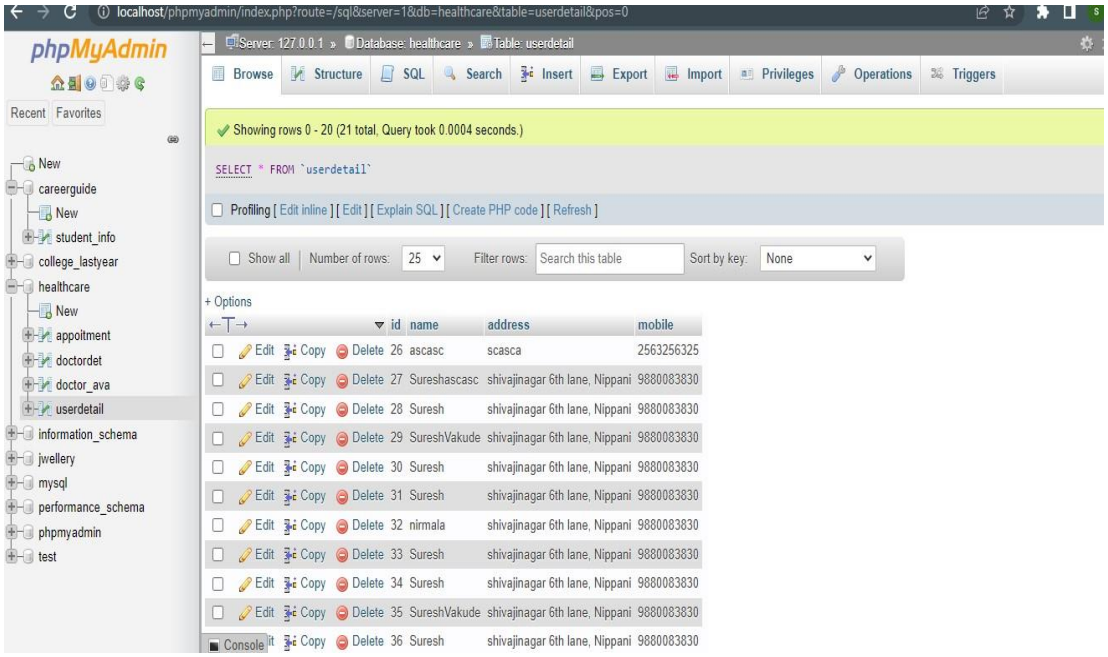




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VI. CONCLUSION

Conversational tools like chatbots are amazing. Here, the application is designed to provide in-depth solutions in a timely manner. It eliminates the load from the answer source by using an expert system to provide the solution directly to the user. The user discovers their ailment remedy with ease. We don't require a doctor's aid or advice for mild illnesses. No charge services. Get assistance if a new illness is discovered. The web interface was created with users' input queries in mind. By providing user protection, character integrity, and retrieving responses in line with the questions, the application is strengthened in terms of security and efficacy.

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